DAY 2 · SESSION 3

Data Protection in Both Directions

Azure Storage, Data Protection, and Resilience at Scale

University of Minnesota

This morning, block by block

01 **Block 1 · 50 min**

Azure Backup architecture and vault selection

02 **Block 2 · 55 min**

Ransomware resilience controls and the live attack demonstration

03 **Break · 15 min**

Coffee and questions

04 **Block 3 · 45 min**

On-premises to Azure and cloud workload protection

05 **Block 4 · 75 min**

Hands-on lab and debrief

Leadership wants to say "we offer immutable data protection" with confidence

The team must define a next-generation protection strategy.

DIRECTION 1

Azure as a target for on-premises backups

DIRECTION 2

Protecting the workloads that will live in Azure

Both flows depend on ransomware resilience controls that make immutability real.

By the end of this session you can

- Describe Azure Backup architecture and choose between Recovery Services vaults and Backup vaults
- Layer immutable vaults, soft delete, and multi-user authorization against ransomware
- Protect on-premises data with the MARS agent and MABS
- Design vaulted and operational backup for Blob, Azure Files, VMs, and managed disks
- Position third-party platforms that use immutable Blob as a target

Picking up from Sessions 1 and 2

01

Session 1

Accounts, tiers, and redundancy decide where data lives and what it costs.

02

Session 2

Immutable Blob storage made data tamper-proof at the storage layer.

03

Today

Backup copies, vault-level immutability, and controls that hold even if an administrator is compromised.

BLOCK 1 · 50 MIN

Azure Backup architecture and vault selection

How backup is built, which vault fits which datasource, and the limits that shape design.

Azure Backup in one picture

Every protected item flows through the same four parts, whatever the workload.

1

Protected source

VM, database, file share, blob, or on-premises server

2

Backup policy

Schedule and retention rules

3

Vault

Stores recovery points and controls operations

4

Management plane

Monitoring, alerts, and reporting at scale

Two vault types, chosen by datasource

ESTABLISHED WORKLOADS

Recovery Services vault

Azure VMs; SQL Server, SAP HANA, and SAP ASE in Azure VMs; Azure Files (snapshots and vaulted backup); on-premises workloads through the MARS agent, MABS, and System Center DPM. Also holds Azure Site Recovery replication data.

NEWER WORKLOADS

Backup vault

Azure Blob (operational and vaulted); Azure Data Lake Storage, meaning hierarchical namespace accounts (vaulted); Azure Disks; Azure Database for PostgreSQL flexible server; Azure Kubernetes Service; Azure Elastic SAN. Azure Cosmos DB vaulted backup is in preview.

Create the vault that matches the datasource. A vault protects resources in its own region, so plan one vault per region per workload class.

Vault limits that shape your design

500

Vaults per subscription

Up to 500 Recovery Services or Backup vaults.

2,000

Datasources per vault

Up to 2,000 across all workloads in a single vault.

1,000

Azure VMs per vault

Up to 1,000, with 250 VM registrations allowed per day.

Plan vault boundaries by workload class, region, and administrative ownership.

Policies, retention, and backup storage redundancy

Policies

Define frequency and daily, weekly, monthly, and yearly retention.

Redundancy first

Vault backup storage redundancy (LRS, ZRS, or GRS) is set before the first item is protected.

Cross-region restore

GRS vaults can enable it to recover in the paired region.

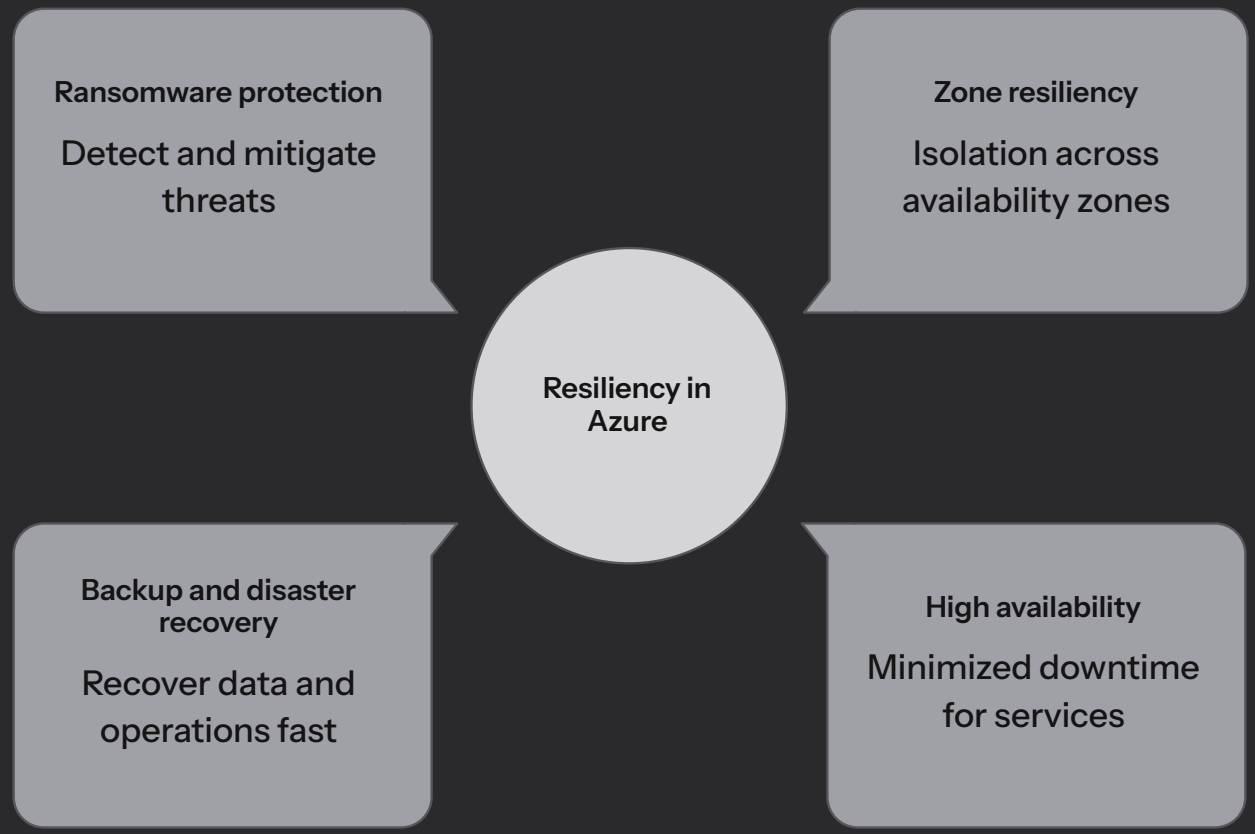
Cost levers

Retention length and redundancy are the two biggest cost levers in backup.

One management plane

Azure Business Continuity Center is now Resiliency in Azure: one place to create and manage vaults, view protection status, and govern backups at scale.

Recovery Services vaults support central monitoring, alerts, metrics, and reports across hybrid assets.



BLOCK 2 · 55 MIN

Ransomware resilience controls and the live attack demonstration

The controls that keep recovery points alive, then watching them hold.

Assume the administrator is compromised

- Ransomware operators target backups first so victims cannot recover
- Controls must hold against someone with valid privileged credentials
- Defense in layers: immutability, soft delete, multi-user authorization, monitoring, least privilege

Immutable vaults: block the operations that lose recovery points

Blocks any operation that could lead to loss of recovery points. Supported on Recovery Services vaults and Backup vaults, for all workloads Azure Backup protects.

Enabled

REVERSIBLE

Can be disabled while you validate.

Locked

IRREVERSIBLE

No one can disable immutability, and WORM storage is enforced.

Immutability applies to vaulted backups. Operational backups of blobs, files, and disks live in your own subscription and are governed by storage-level controls instead.

How long immutability applies

OPTION A

Based on backup policy

Immutability matches each protected item's policy retention.

OPTION B

Specific duration

Immutable for a set number of days, then points live on per policy without immutability.

Policy retention cannot be reduced below that duration.

Time-based immutability is supported for Recovery Services vaults.

WORM storage behind locked immutability

01

Locked immutability is enforced everywhere

Enabling and locking immutability is generally available in all Azure public and US government regions, for every workload Azure Backup protects.

02

WORM-backed storage: Backup vaults

Generally available in all Azure public regions for AKS, PostgreSQL flexible server, and Cosmos DB backups.

03

WORM-backed storage: Recovery Services vaults

Generally available in 20 regions, including West US 3, Japan East, Poland Central, and Switzerland North; not yet in the central or eastern US regions. Covers Azure VMs, SQL and SAP HANA in VMs, Azure Files, MARS, MABS, and DPM.

04

Everywhere else

The backup service enforces locked immutability itself, and backups move to WORM-backed storage automatically when it arrives in that region, with no action and no data movement.

Check the support matrix for your region and workload before you say "WORM" to leadership; say "locked immutable vault" where WORM storage has not yet arrived. Both block the same operations.

Soft delete and enhanced soft delete

Soft delete

Deleted backup data is retained for a period so it can be recovered.

Enhanced soft delete

Configurable retention period and an always-on setting that cannot be turned off once enabled.

- Soft delete is on by default: deleted backup data is kept 14 days at no cost, protecting against accidental and malicious deletion
- Enhanced soft delete lets you set retention up to 180 days and make soft delete always-on, which cannot be turned off once enabled; soft-deleted data still incurs storage cost beyond the free window

Multi-user authorization with Resource Guard

Second authorization

Resource Guard gates critical operations on a vault.

Protected operations

Disabling soft delete, reducing retention, and removing protection.

Separate ownership

Place it in a separate subscription or tenant owned by a different team.

No single actor

The vault administrator alone cannot perform a protected operation.

See it, then contain it: monitoring and least privilege

See it

- Built-in alerts for security-relevant events such as deletion of backup data and disabling of protections
- Route alerts through Azure Monitor action groups to the right team

Contain it

- Use built-in roles: Backup Reader, Backup Operator, Backup Contributor
- Grant the narrowest role, at the narrowest scope, for the shortest time

Layered defenses at a glance

Attack path	Control that stops it
Delete recovery points	Immutable vault
Reduce policy retention	Immutable vault and multi-user authorization
Disable soft delete	Enhanced soft delete always-on and multi-user authorization
Purge immutable backup container	Locked time-based immutability on Blob
Act unnoticed	Alerts and audit through Azure Monitor
Delete the storage account holding backup images	Locked time-based immutability on Blob (an unlocked policy does not block account deletion)

Demonstration: a ransomware-style attack, defeated live

01 Delete backup data from an immutable Recovery Services vault

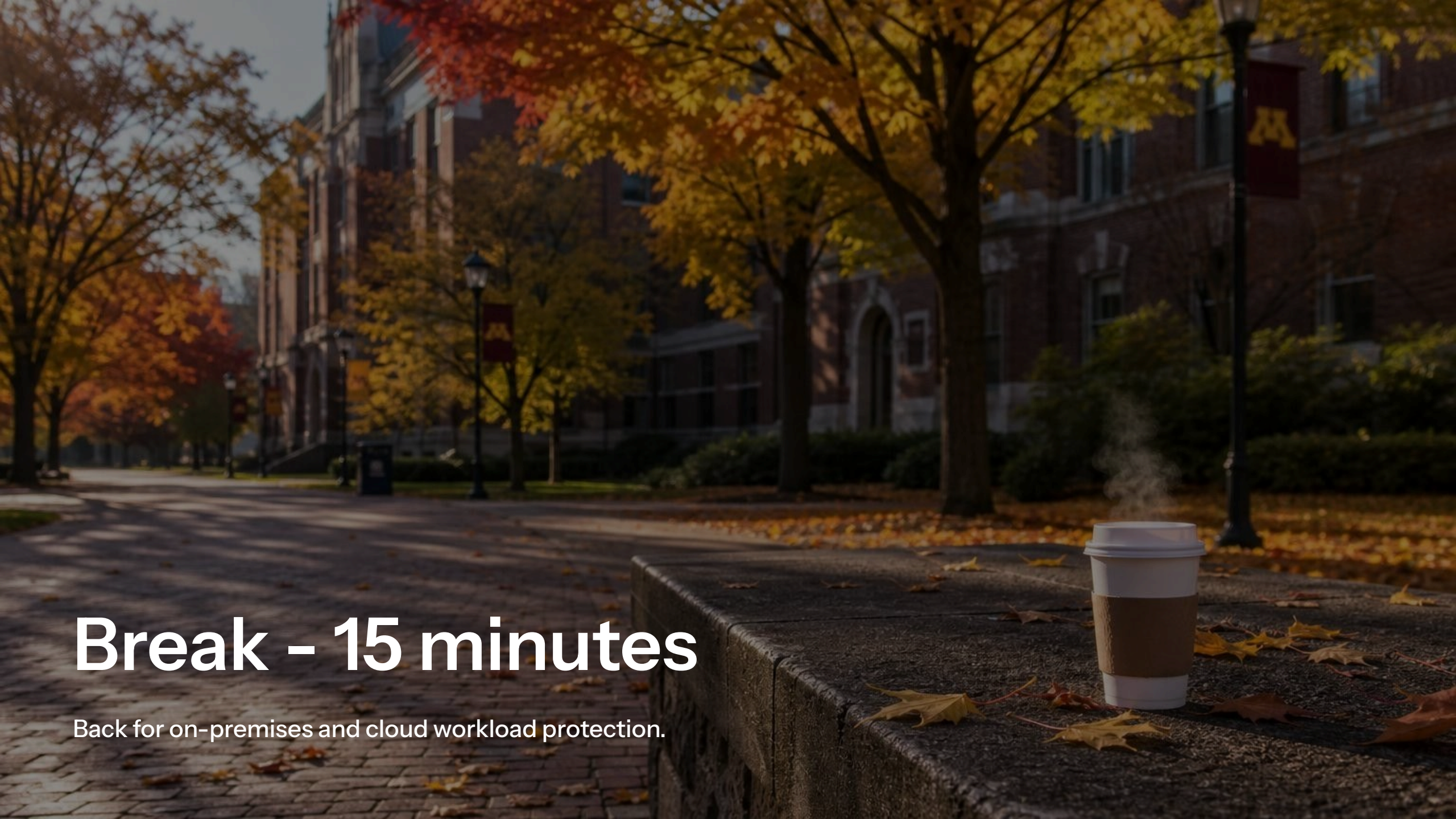
02 Reduce retention on a backup policy

03 Disable soft delete

04 Purge an immutable blob container holding on-premises backup images

05 Delete the storage account that holds the on-premises backup images

Five attack paths, five platform-enforced refusals, each with its alert and audit entry.



Break - 15 minutes

Back for on-premises and cloud workload protection.

BLOCK 3 · 45 MIN

On-premises to Azure and cloud workload protection

Both directions: campus servers into Azure, and the workloads already running there.

MARS agent: files, folders, volumes, and system state

On-premises

Windows files and folders, entire volumes, and system state to Azure.

Inside Azure VMs

Specific files and folders.

Efficient transfer

Initial backups compressed; later ones send only incremental changes.

Offline seeding

Azure Data Box for large initial backups.

MABS: application-aware on-premises protection

Microsoft Azure Backup Server protects workloads such as SQL Server, Exchange, SharePoint, Hyper-V, and VMware VMs.

Two tiers

Local disk for fast restores, then a Recovery Services vault for offsite retention.

Bandwidth throttling

Protects production network traffic.

Support matrix

Check current versions and workloads.

Encryption passphrase custody is your responsibility

- MARS and MABS encrypt data with a passphrase you set before it leaves the server
- Microsoft cannot recover lost passphrases; without it, the backups cannot be restored
- Store the passphrase in Azure Key Vault or a secure location outside the protected servers
- Document custody, access, and rotation in the recovery runbook

Blob backup: operational or vaulted

CONTINUOUS

Operational

Retention up to

360 days

Local protection in the source account using point-in-time restore, soft delete, change feed, and versioning.

Restores go to the source account only and can overwrite data.

PERIODIC

Vaulted

Retained up to

10 years

Offsite copy in the Backup vault on a schedule.

Operational backup relies on point-in-time restore, change feed, and versioning, which hierarchical namespace accounts do not support. For Data Lake Storage accounts use vaulted backup for Azure Data Lake Storage, generally available, with auto-protection for up to 1,000 containers per account. Immutable vault settings apply to the vaulted tier only.

Protecting Azure Files, VMs, and managed disks

Azure Files

Share snapshots managed by a Recovery Services vault, plus vaulted backup for an offsite copy.

Azure VMs

Application-consistent backups through the VM backup extension in a Recovery Services vault.

Managed disks

Incremental snapshot-based backup managed by a Backup vault.

Enable container soft delete alongside blob operational backup: deleted containers cannot be restored by operational backup.

A backup is only as good as its last tested restore

1
Schedule →
Restore tests per workload class.

2
Restore safely →
To an alternate location, never over production.

3
Validate →
Data integrity and application function.

4
Record
Results as evidence for audit and leadership.

Third-party backup platforms with immutable Blob targets

How it works

Enterprise backup platforms can write to Blob containers protected by time-based immutability or legal holds.

Object lock semantics map to version-level WORM, which requires blob versioning.

When to use it

Cool and Archive reduce capacity cost, but watch minimum retention and retrieval charges for backup chains.

Use when the platform already protects the on-premises estate and needs an immutable cloud copy.

Keep the protected copy outside the blast radius

- Vaulted backup data lives in a Microsoft-managed subscription and tenant, isolated from your environment and a compromised administrator
- A vault must share the datasource's region, and for most workloads its subscription; cross-subscription backup for Azure VMs and Azure Files is in preview - plan vault placement per workload
- Put the Resource Guard, not necessarily the vault, in a separate subscription or tenant owned by a different team
- Different administrators and RBAC scopes for production and backup: Backup Operator and Backup Reader for daily work, Backup Contributor only where needed
- Enforce protection standards with Azure Policy: require backup, immutability, and soft delete
- Consider data residency when choosing vault regions and cross-region restore

Which vault, which control, which workload

Workload	Vault or mechanism	Key controls
On-premises files and system state	Recovery Services vault via MARS	Immutable vault, passphrase custody
On-premises applications	MABS to Recovery Services vault	Immutable vault, MUA
Azure VMs	Recovery Services vault	Immutable vault, enhanced soft delete, MUA
Azure Files	Recovery Services vault	Snapshots plus vaulted backup
Azure Blob	Backup vault, operational and vaulted	Immutable vault and MUA for the vaulted tier; soft delete and versioning for the operational tier.
Third-party backup images	Immutable Blob container	Locked time-based retention
Azure Data Lake Storage (HNS accounts)	Backup vault, vaulted only	Immutable vault, MUA

BLOCK 4 · 75 MIN

Hands-on lab and debrief

Build the controls yourself, then test whether they hold.

Lab: an immutable Backup vault with multi-user authorization

01

Deploy

A Backup vault with immutability enabled.

02

Guard

Create a Resource Guard in a separate resource group and link it.

03

Test

Attempt a protected operation; confirm authorization is required.

04

Protect

A storage account with vaulted Blob backup.

05

Restore

To an alternate location.

Lab notes

Stay unlocked

Keep immutability enabled but unlocked so lab resources can be cleaned up.

Start early

Begin the vaulted backup first: the first backup takes time to complete.

Check as you go

Use the validation checkpoints in the lab guide after each step.

Debrief questions

1. Which control would have stopped the attack you worry about most?

2. Who in the university should own the Resource Guard?

3. What would you tell your vice president "immutable" means now?

Key takeaways and what comes next

- Choose the vault by datasource; set redundancy before protecting items
- Immutability, enhanced soft delete, and multi-user authorization together resist a compromised administrator
- Protect both directions, and keep the copy outside the primary blast radius

Materials

Demo script, lab guide, protection pattern reference, executive briefing one-pager

Next

Resilience, disaster recovery, and multi-region design

Capstone input: Immutable backup target for on-premises protection